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Modeling and Control of Cable-Driven UAV Systems for Collaborative Load Manipulation

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Abstract

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Notations

$F_0(O, x_0, y_0, z_0)$	Inertial reference world frame
$F_p(P, x_p, y_p, z_p)$	Platform reference frame at its center of mass
$F_i(A_i, x_i, y_i, z_i)$	Quadrotor reference frame at cable connection
$F_{C_i}(A_{C_i}, x_{C_i}, y_{C_i}, z_{C_i})$	Quadrotor reference frame attached to the quadrotor at its center of mass
$F_k(A_k, x_k, y_k, z_k)$	Ground vehicle reference frame

Abbreviations

ACT	Aerial Cable Towed
ACTR	Aerial Cable Towed Robot
ACTS	Aerial Cable Towed System
AW	Available Wrench set
CDPR	Cable-Driven Parallel Robot
DKM	Direct Kinematic Model
DoF	Degree of Freedom
IKC	Inverse Kinematic Controller
INDI	Incremental Nonlinear Dynamic Inversion
MPC	Model Predictive Control
NMPC	Nonlinear Model Predictive Control
OCP	Optimal Control Problem
ORCA	Optimal Reciprocal Collision Avoidance
rAW	Radius of the Available Wrench
SAW	Static Available Wrench set
SW	Static workspace
UAV	Unmanned Aerial Vehicle
UGV	Unmanned Ground Vehicle
VIC	Variable Impedance Control
WFW	Wrench Feasible Workspace

List of Figures

1.1 Graphical representation of the platform 10
1.2 Forces acting on the ACTS 12

2.1 Possible UAVs connection to the payload 14
2.2 Possible UGVs connection to the payload 14
2.3 One drone system connected to the ground through a cable 15
2.4 Simplified ACTS with one drone and one ground anchor 16
2.5 Tension directed arbitrary 17
2.6 Payload connected through two cable to the ground 17
2.7 Over actuated ACTS with 6 ground cables and 3 UAVs 21
2.8 High level control scheme 22
2.9 Gogh Stewart schematics [1] 23

3.1 Performance indices for 33
3.2 Possible 6 UGVs connections 34
3.3 Possible 3 UAVs connection with one cable 34

List of Tables

3.1 Comparative performance indices and tracking errors for Stewart baseline configuration across Scenarios A and B. 32
3.2 Ground Cable Routing and Layout Performance Analysis 35

A.1 Classification of Singularity Types in Cable-Driven Parallel Systems . . . 40

Contents

1 Kinemato-static model **9**
1.1 Kinematic model 9
1.2 Static model 11
1.3 Kinemato-static model 12

2 ACTS starting configuration **13**
2.1 Connections to the payload 13
2.2 Force Control on point-mass payload 15
2.2.1 Case: one drone connected to the ground through a cable 15
2.2.2 Case: payload connected to the ground and a drone 16
2.2.3 Case: payload connected to the ground through two cables 17
2.3 Performance Indices 19
2.4 Architecture Selection Strategy 21
2.5 High level architecture 22
2.5.1 Quadratic Approximation of the UAV Disposition Problem 25

3 Experiments **27**
3.1 Experimental methodology and evaluation framework 29
3.2 Stewart platform baseline 30
3.2.1 Numerical Verification of the Implementation 30
3.2.2 Comparison τ_{min} and τ_{opt} 31
3.3 Ground cable routing 32
3.4 Ground anchor placement 36
3.5 Drone hook placement 36

4 Conclusions **37**

Appendix **39**

A Singularities and Performance Indices **40**

Chapter 1
Kinemato-static model

A kinematostatic model is a model that combines kinematics (motion and geometry) with statics (forces and equilibrium) to describe how a mechanical system moves and how forces are transmitted through it.

1.1 Kinematic model

Given the payload twist ${}^0\xi = \begin{bmatrix} {}^0v_p & {}^0\omega_p \end{bmatrix}^\top$ in the inertial reference frame $< 0 >$, we can write the position and velocity of the attachment point B_i (resp. B_k) of the aerial vehicle (resp. ground) in $< 0 >$.

$$ {}^0\boldsymbol{r}_i = {}^0\,\boldsymbol{p} + {}^0_PR\,{}^P\boldsymbol{b}_i \tag{1.1}$$

where b_i is the attachment point on payload.

$$ {}^0\dot{\boldsymbol{r}}_i = {}^0\,\boldsymbol{v}_p + {}^0\,\boldsymbol{\omega}_p \times \left({}^0_PR\,{}^P\boldsymbol{b}_i\right) \tag{1.2}$$

For definition, the cable vector is $\boldsymbol{l}_i = \boldsymbol{a}_i - \boldsymbol{r}_i = A_i - B_i$ with $\boldsymbol{a}_i$ being the anchor position. Under a centralized controller, the positions of all UAV A_i and UGV A_k are known, which directly determin the anchor vector $\boldsymbol{a}_i$. Furthermore, he payload's position and orientation are assumed to be known.ogether with the known locations of the attachment points relative to the payload, this information allows us to calculate the cable vectors $\boldsymbol{l}_i$.

The final form of cable vector variation is given by applying Equation 1.2 in its definition:

$$ \dot{\boldsymbol{l}}_i = \dot{\boldsymbol{a}}_i - \dot{\boldsymbol{r}}_i = \dot{\boldsymbol{a}}_i - \frac{d}{dt}\left(\boldsymbol{p} + {}_PR\,{}^P\boldsymbol{b}_i\right) = \dot{\boldsymbol{a}}_i - \left(\boldsymbol{v}_p + \boldsymbol{\omega}_p \times \left({}_PR\,{}^P\boldsymbol{b}_i\right)\right) \tag{1.3}$$

Each $\dot{l}_i$ corresponds to the projection of the relative velocity of B_i along the cable direction

$$\begin{aligned} \dot{l}_i &= u_i^\top \left(\dot{\boldsymbol{a}}_i - \left(\boldsymbol{v}_p + \boldsymbol{\omega}_p \times \left({}_PR\,{}^P\boldsymbol{b}_i\right)\right)\right) \\ &= u_i^\top \dot{\boldsymbol{a}}_i - \begin{bmatrix} u_i^\top & \left({}_PR\,{}^Pb_i \times u_i\right)^\top \end{bmatrix} \xi \\ &= J_{a,i}\,\dot{a}_i - J_{p,i}\,\xi, \end{aligned} \tag{1.4}$$

This gives us the kinematic model that relates the variation of cable length vector $\boldsymbol{\rho}$ with the change in position of the anchors and the payload twist:

$$ \dot{\rho} = J_a\dot{a} - J_p\xi = J_a \begin{bmatrix} \dot{a}^{(a)} \\ \dot{a}^{(g)} \end{bmatrix} - J_p\xi \tag{1.5}$$

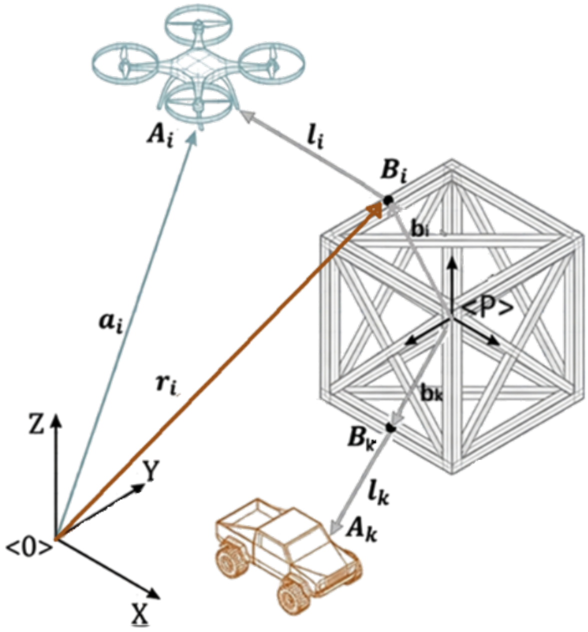

Figure 1.1: Graphical representation of the platform

Let $m = n_a + n_g$ be the total number of winches (or cables) and $u_i = l_i/\|l_i\|$ the unit cable direction , then:

- $\dot{\rho} = \begin{bmatrix} \dot{l}_1 & \dot{l}_2 & \dots & \dot{l}_m \end{bmatrix}^\top$ is the cable length rate (dimension $[m \times 1]$).
- J_a is the Jacobian relating the anchor velocities to the cable-length rates $[m \times 3m]$. In particular, each cable depends only on its anchor velocity.

$$J_a = \begin{bmatrix} u_1 & 0 & \cdots & 0 \\ 0 & u_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{n_a+n_g} \end{bmatrix}, \qquad u_j = \frac{(a_j - r_j)}{\|a_j - r_j\|}. \tag{1.6}$$

- J_p is the Jacobian relating the payload twist to the cable-length $[m \times 6]$

$$J_p = \begin{bmatrix} u_1^\top & ({}_pR\,{}^pb_1 \times u_1)^\top \\ \vdots & \vdots \\ u_m^\top & ({}_pR\,{}^pb_m \times u_m)^\top \end{bmatrix} \tag{1.7}$$

Note 1. Equation 1.4 contains m equations in 6 unkown. I need at least six independent cable directions (rank$(J_p) = 6$).

The control allocation matrix maps the actuator inputs (cable tension vector) to the system (payload) wrench. For this definition $J_p^\top$ is the allocation matrix.

$$W_p = J_p^\top \tau \tag{1.8}$$

Let W_p^* being the desired payload wrench, the control allocation problem to solve is

$$J_p^\top \tau = W_p^*, \qquad \text{s.t.} \tau_i > 0 \tag{1.9}$$

1.2 Static model

Let $\tau = [\tau_{uav}^\top, \tau_{ugv}^\top]^\top \in \mathbb{R}^{m\times 1}$ the cable tension vector, then

$$\begin{aligned}\underbrace{\tau^\top \dot{\rho}}_{\text{total cable power}} &= \underbrace{\tau^\top J_a \dot{a}}_{\text{UAVs and UGVs mechanical power}} - \underbrace{\tau^\top J_p \xi}_{\text{payload power}} \\ &= (J_a^\top \tau)^\top \dot{a} - (J_p^\top \tau)^\top \xi \\ &= F_a^\top \dot{a} - W_p^\top \xi\end{aligned} \tag{1.10}$$

Each cable i applies a force on the payload $F_{a,i}$ and an equal and opposite force on the UAV (resp. UGV). We can write balance equation on our elements:

Payload Balance

$$\underbrace{W_p}_{J_p^\top \tau} + \underbrace{W_g}_{\begin{bmatrix}0\\0\\-m_p g\\0\\0\\0\end{bmatrix}} + W_{contact} + \underbrace{W_{wind}}_{\text{let } = 0} = 0 \tag{1.11}$$

where since each cable produces a wrench on the payload, the contribution given by the i-th cable is:

$$\underbrace{w_{p,i}}_{6\times 1} = \underbrace{J_{p,i}^\top}_{6\times 1} \underbrace{\tau_i}_{1\times 1} = \begin{bmatrix} u_i^\top & ({}_P R\, {}^P b_i \times u_i)^\top \end{bmatrix}^\top \tau_i = \begin{bmatrix} \tau_i u_i \\ ({}_P R\, {}^P b_i \times \tau_i u_i) \end{bmatrix} \tag{1.12}$$

and $W_{contact} = 0$ when there is no contact with the environment.

UAV balance

$$-F_{a,i} + F_{prop,i} + \underbrace{F_{g,i}}_{\begin{bmatrix}0\\0\\-m_i g\end{bmatrix}} = 0 \tag{1.13}$$

with ${}^i F_{prop} = \begin{bmatrix} 0 & 0 & F_{prop,z,i} \end{bmatrix}^\top$ and $F_{a,i} = J_{a,i}^\top \tau_i = \tau_i u_i$.

UAVs must always generate sufficient thrust to maintain flight and ensure their own stability. As a result, only a portion of their available thrust can be allocated to payload manipulation, which reduces the effective force that can be transmitted through the cables.

UGV balance

$$-F_{a,j} + \underbrace{F_{ground,j}}_{\begin{bmatrix}F_{frict,x,j}\\F_{frict,y,j}\\F_{norm}\end{bmatrix}} + \underbrace{F_{g,j}}_{\begin{bmatrix}0\\0\\-m_j g\end{bmatrix}} = 0 \tag{1.14}$$

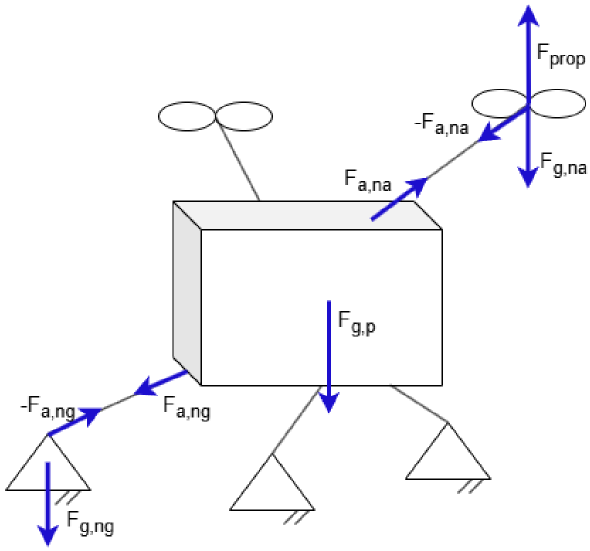

Figure 1.2: Forces acting on the ACTS

with $\|F_{frict}\| \leq \mu F_{norm}$ and $F_{norm} = m_j g - F_{a,z,j}$ and $F_{a,j}$ is obtained through Equation 1.15.

Similarly, UGVs are subject to ground interaction constraints. In particular, they must satisfy friction limits to remain stationary when no motion is commanded. This imposes bounds on the admissible cable forces, as excessive tension may cause slipping. By Newton's third law, the force the cable exerts on the drone $F_{a,i}$ (resp. on the UGV $F_{a,j}$) is equal and opposite to the force it exerts on the payload. Since u_i points from B_i toward A_i, the cable pulls the drone with

$$F_{a,i} = J_{a,i}^\top \tau_i = \tau_i u_i \tag{1.15}$$

As a consequence, the total cable force on the payload is:

$$F_p = -\sum_{i=1}^{n_a} F_{a,i} - \sum_{j=n_a+1}^{n_a+n_g} F_{a,j} \tag{1.16}$$

where the payload wrench is $W_p = \begin{bmatrix} F_p & M_P \end{bmatrix}^\top$

1.3 Kinemato-static model

Writing Equation 1.11, 1.13 and 1.14 in term of τ, we obtain:

$$W_{sys} = \begin{bmatrix} W_p \\ F_a \end{bmatrix} = \begin{bmatrix} J_p^\top \\ J_a^\top \end{bmatrix} \tau = G\tau = \begin{bmatrix} -W_g - W_{contact} - W_{wind} \\ F_{prop} + F_{ground} + F_g \end{bmatrix} \tag{1.17}$$

where $F_{prop,j} = 0$ for the UGVs and $F_{ground,i} = 0$ for the UAVs. Then W_{sys} is $(6+3m) \times 1$ and

$$G\tau + \begin{bmatrix} W_g + W_{contact} + W_{wind} \\ -F_{prop} - F_{ground} - F_g \end{bmatrix} = 0 \tag{1.18}$$

Chapter 2

ACTS starting configuration

The ACTS consists of at least one UAV, required as the payload needs to be lifted from the ground, and one UGV, included for safety and operational constraints. To fully control the payload in space, we need at least six cables

2.1 Connections to the payload

The configuration of the UAV connections is crucial as it directly affects how forces are applied to the payload. UAVs are responsible for generating the lifting force required to lift the payload. Consequently, the number of UAVs depends strongly on the payload mass. However, increasing the number of UAVs leads to higher power consumption and greater system complexity. So, adding more UAVs does not necessarily improve performance. Additionally, each UAV can be connected to the payload using either a single cable or multiple cables. The choice of configuration affects controllability, wrench feasibility, robustness, and energy efficiency, and should be evaluated accordingly. A practical compromise is to use three UAVs connected to the payload at distinct attachment points, providing a balance between the metrics just mentioned (Figure 2.1).

In contrast, UGVs do not contribute to lifting the payload, meaning the same stability arguments do not apply. If we treat the UGVs as fixed anchor points, connecting a single UGV to multiple hook points on the payload simplifies to a standard fixed-anchor scenario. Because each actuator operates independently, this configuration adds no extra mathematical complexity. However, if the UGVs are actively moving, multi-cable connections require precise winch coordination to ensure all cables remain taut at all times. In Figure 2.2, possible payload hook attachment can be found.

Since the system is designed to be redundant, the overall number of UAVs and UGVs must be carefully selected.

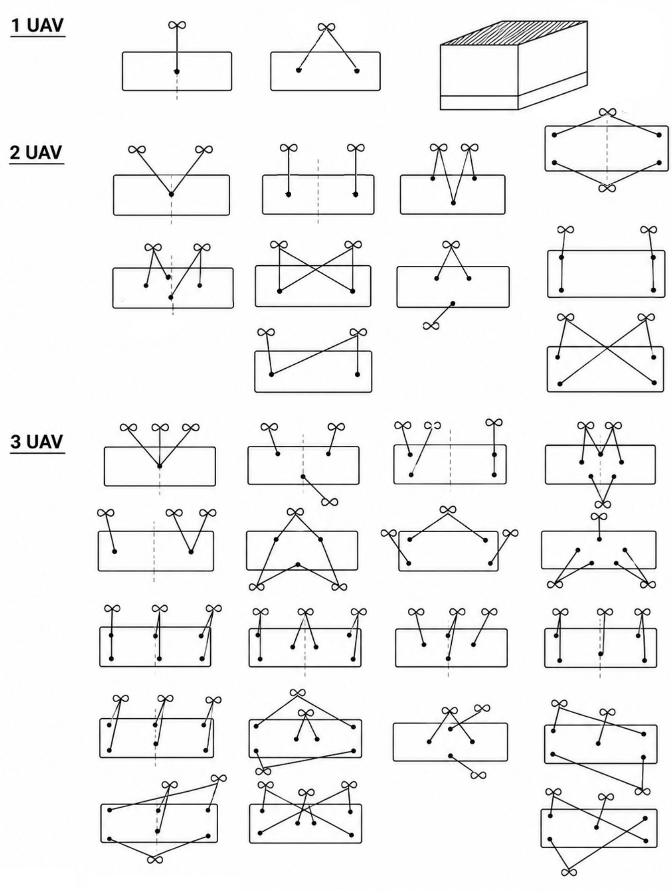

Figure 2.1: Possible UAVs connection to the payload

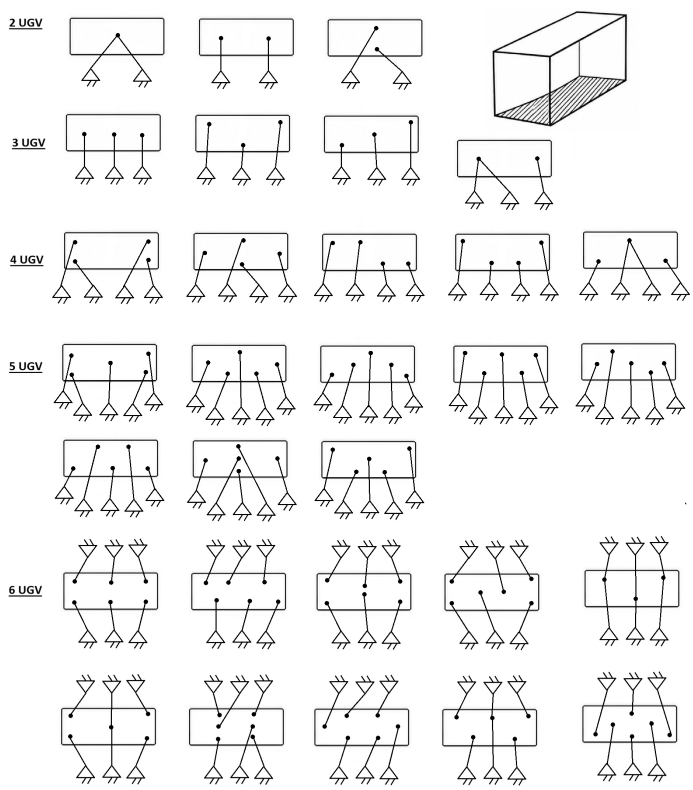

Figure 2.2: Possible UGVs connection to the payload

2.2 Force Control on point-mass payload

Before going into details on the full consideration, let's analyze some simpler configurations.

2.2.1 Case: one drone connected to the ground through a cable

The goal is to design the drone force F_{prop} such as that the ground tension is controlled. Since only a tension in the cable direction is imposable, an arbitrary force f^* at the ground anchor can be achieved only by moving the drone, changing the cable direction u. Called a the drone's position and b the attachment point to the ground:

$$u = \frac{a-b}{\|a-b\|} \tag{2.1}$$

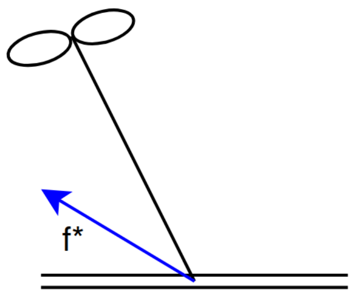

Figure 2.3: One drone system connected to the ground through a cable

Option 1

Defined the desired cable tension τ^* and the desired cable direction u^* as:

$$\tau^* = \|f^*\| \tag{2.2}$$

$$u^* = \frac{f^*}{\|f^*\|} \tag{2.3}$$

the ground tension is obtained using the geometric constraint which relates the anchor point to the drone position:

$$a^* = b + l\, u^* \tag{2.4}$$

where l is the cable length. Equation 2.4 is then used as a reference to compute the propeller force.

$$F_{prop} = m_i g e_3 + K_p(a^* - a) + K_d(\dot{a}^* - \dot{a}) + \tau^* u \tag{2.5}$$

Option 2

Given the drone Newton's Second Law described in Equation 1.13, we are able to estimate the cable force $\hat{f} = F_{a,i}^{est}$

$$m_i \ddot{a}_i = F_{prop,i} - \hat{f} + F_{g,i} \qquad \rightarrow \qquad \hat{f} = F_{prop,i} + F_{g,i} - m_i \ddot{a}_i \tag{2.6}$$

where all the forces are in the world frame and $\ddot{a}_i$ is the drone linear acceleration. Using Equation 1.13, we would like to apply a propulsion force equal to

$$F^*_{prop,i} = f^* - F_{g,i} \tag{2.7}$$

Finally, the closed loop equation is:

$$F_{prop,i} = F^*_{prop,i} + K_p e + K_d \dot{e}, \quad \text{with } e = f^* - \hat{f} \tag{2.8}$$

The result is given to the attitude controller, which aligns the body z-axis with the commanded thrust vector.

2.2.2 Case: payload connected to the ground and a drone

In this case, we are working with the simplified case of an ACTS composed by a drone, a ground anchor and a point-mass payload. Called with subscript 1 the cable that connects the payload to the drone and with 2 the one that connects it to the ground, for definition the vector u_i, $i = 1 \div n$ is given by

$$u_i = \frac{a_i - p}{\|a_i - p\|} \tag{2.9}$$

being a_i and p respectively the actuator and the payload position.

Payload position control

The goal is to design $F_{prop,1}$ such that the payload reaches a desired position ($p \to p^*$), which is reachable for the length of the cable.

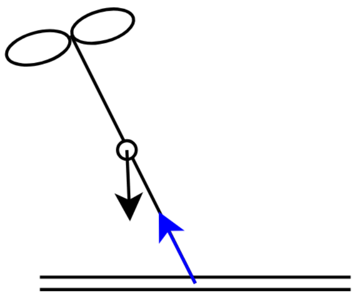

Figure 2.4: Simplified ACTS with one drone and one ground anchor

The payload dynamics is given by:

$$\begin{aligned} m_p \ddot{p} &= \tau_1 u_1 + \tau_2 u_2 - m_p g e_3 \\ &= F_{a,1} + F_{a,2} + F_g \qquad \rightarrow \qquad F_p = m_p(\ddot{p} + g e_3) \\ &= F_p + F_g \end{aligned} \tag{2.10}$$

The desired payload force is defined using an impedance controller:

$$F^*_p = m_p(\ddot{p}^* + g e_3) + K_p(p^* - p) + K_d(\dot{p}^* - \dot{p}) \tag{2.11}$$

with $\dot{p}^* = 0$ and $\ddot{p} = 0$ as we required the payload to maintain the position once it is reached. The remaining force to be generated by cable 1 is:

$$F^*_{a,1} = F^*_p - F_{a,2} \tag{2.12}$$

where $F_{a,2} = \tau_2 u_2$. This brings us back to the situation in Section 2.2.1, where $F_{a,1}$ is f^*.

Tension and orientation control

The goal can also be design $F_{prop,1}$ such that an arbitrary force f^* on the ground can be applied ($F_2 \rightarrow f^*$). In other words, $u_2 \rightarrow u_2^*$ and $\tau_2 \rightarrow \tau_2^*$. This case can be interesting to study as in a redundancy system situation, it may be helpful to add additional constraint in one of the cables. The only difference from the previous case is that the desired position for the payload p^* is not an input anymore, but obtained such that the 2nd cable is aligned along the desired direction given by f^* (Equation 2.2 and 2.3). Then:

$$p^* = a_2 - l_2 u_2^* \tag{2.13}$$

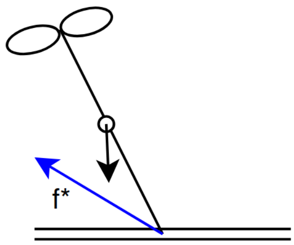

Figure 2.5: Tension directed arbitrary

2.2.3 Case: payload connected to the ground through two cables

The ACTS consists of a point-mass payload connected to the ground through two cables and to an UAV. The enumeration is shown in Figure 2.6.

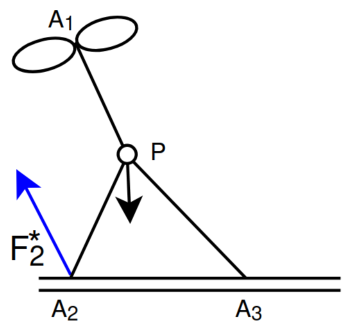

Figure 2.6: Payload connected through two cable to the ground

Payload position control

If the input is the desired payload position p^*, we find ourselves in three situations:

1. p^* lies precisely on the intersecting boundary arcs dictated by the maximum lengths of the ground cables;

2. p^* is out of reach;

3. p^* is inside the triangle A_1A_3P, resulting into slack cables

In the first case, p^* is used to compute the desired payload force F_p^* (Equation 2.11). Using the simplified payload balance equation related to the forces acting on the payload (Equation 1.16), we obtain the following.

$$F_{a,1}^* = F_p^* - \sum_{i=2}^{n} F_{a,i} \tag{2.14}$$

which brings us back to the previous case.

Because the payload is physically constrained at a boundary where both ground cables are stretched straight, any movement or external disturbance is perfectly resisted by the tension of the cables.

In the second case, since the ground cables physically block the payload from reaching p^*, a steady error $e = (p^* - p) \neq 0$ builds up. The actual tension will be directly proportional to the control gains and the distance of the unreachable input:

$$\sum F_{virtual} \propto (p^* - p_{geom}) \quad \Longrightarrow \quad \sum F_{virtual} = K(p^* - p_{geom}) \tag{2.15}$$

where p_{geom} is the closest reachable point that lies on the workspace boundary. In this case, we may impose a controlled tension on the ground cables directed along $(p^* - p_{geom})$ to ensure the drone continues to pull toward the desired target:

$$F_{a,1}^* = F_p^* - \sum_{i=2}^{n} F_{a,i} + F_{virtual} \tag{2.16}$$

In the last case, the distance from the anchors to the payload is less than the cable length, resulting into slack cables as the latter cannot push. Mathematically $\tau_2 = 0$ and/or $\tau_3 = 0$. Then, $F_{a,1}^* = F_p^*$

Desired tension in the cable on the ground

The desired variable to control is the tension of one of the cables $F_{a,2} \to f^*$.

$$F_{a,1}^* = F_p - f^* - F_{a,2} \tag{2.17}$$

where $F_p = m_p(\ddot{p} + ge_3)$ and $F_{a,2} = \tau_2 u_2$.

Desired tension difference between the cables on the ground

The desired variable to control a linear combination of the two $F_{a,2} + \alpha F_{a,3} \to f^*$.

$$F_{a,1} = F_p - F_{a,2} - F_{a,3} \quad \Longrightarrow \quad \begin{aligned} F_{a,1}^* &= F_p - (f^* - \alpha F_{a,3}^* + F_{a,3}^*) \\ &= F_p - f^* - (1-\alpha) F_{a,3} \end{aligned} \tag{2.18}$$

For the specific case where $F_{a,2} - F_{a,3} \to f^*$:

$$F_{a,1}^* = F_p - f^* - 2F_{a,3}^* = F_p - f'^* \tag{2.19}$$

2.3 Performance Indices

In the context of the ShieldBot project, we are working with two questions:

1. Which attachment geometry gives the best ability to generate the required payload wrench over the entire mission workspace?

2. Given the chosen architecture, where should the actuators (UAVs and anchor points in the ground) be positioned and what cable lengths should be used during operation?

In both cases, we need an objective measurement that guides us in the selection.
Every configuration to be eligible needs to satisfy feasibility metrics first, such as [2]:

- Wrench closure conditions (WCC), which guarantees that the payload is fully constrained and arbitrary wrenches can be generated. Mathematically: $\operatorname{rank}(J_p) = 6$ and positive tensions exist.

- Wrench Feasible Condition (WFC), which checks whether tensions remain in the feasible bound $0 \le \tau_{\min} \le \tau \le \tau_{\max}$, included UAV thrust limits, cable strength limits and UGV friction limits in a future where the ground anchor point will be movable.

- Interference-Free Condition (IFC), ensuring that there is no cable rubbing and no collision with payload edges. Mathematically:

$$\begin{aligned} d_{ij} &\ge d_{\text{safe}}, \quad \text{where } d_{ij} = \operatorname{dist}(\text{cable } i, \text{cable } j) \qquad \forall i, j \\ d_i &\ge d_{safe}, \quad \text{where } d_i = \min_{s\in[0,1]} \left(\operatorname{dist}\left(r_i + s(a_i - r_i), P\right)\right) \qquad \forall i \end{aligned} \tag{2.20}$$

  where P is the payload geometry and d_{safe} is a clearance threshold.

check that this is a right conclusion

However, assuming that all cable attachment points are on the same face of the payload, the only feasible interference between the latter and one of the cables is a near-tangent cable that touches that face. This requirement will be assumed to always be met, as this study places the ACTS in a scenario where this corner case is rarely encountered.

After passing the feasibility filters, we need a way to assign a score to each hook configuration.

Capacity margin or minimum degree of constraint satisfaction

The capacity margin is a robustness index that is used to analyze the degree of feasibility of a configuration. It is defined as the shortest signed distance from the task wrench to the boundary of the available wrench set (AWS). Mathematically, it is the minimum of the distances of each one of the n vertex $w_{e,j}$ to each one of the p facet $(\mathbf{a}_l, b_l)$ of the AWS.

$$\gamma = \min_{j=1\div n} \min_{l=1\div p} \gamma_{j,l}, \quad \text{where } \gamma_{j,l} = \frac{b_l - w_{e,j}^{\top}\mathbf{a}_l}{\|\mathbf{a}_l\|_2} \tag{2.21}$$

where $\mathbf{a}_l$ is the normal vector to the l-th facet and b_l is its offset [3].

Worst-Case Capacity Margin

Additionally, we can define the worst-case capacity margin as how far the system remains from actuator saturation in the most demanding operating condition. Mathematically:

$$\zeta = \min \left(1 - \max_i \frac{\tau_i}{\tau_{i,max}} \right) \tag{2.22}$$

For each drone, the maximum available cable tension $t_{i,max}$ is based on its maximum thrust $f_{i,max}$ and its current orientation relative to the cable [4]

$$t_{i,max} = m_i \boldsymbol{u}_i^T (\boldsymbol{g} - \ddot{\boldsymbol{x}}_i) + \sqrt{f_{i,max}^2 + m_i^2 \left((\boldsymbol{u}_i^T (\ddot{\boldsymbol{x}}_i - \boldsymbol{g}))^2 - \ddot{\boldsymbol{x}}_i^T \ddot{\boldsymbol{x}}_i - \boldsymbol{g}^T \boldsymbol{g} \right)} \tag{2.23}$$

Considering only quasi-static motion, this can be reduced to the following, where $u_{iz} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \boldsymbol{u}_i$ and $g = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \boldsymbol{g}$.

$$t_{i,max} = m_i g u_{iz} + \sqrt{f_{i,max}^2 + m_i^2 g^2 \left(u_{iz}^2 - 1 \right)} \tag{2.24}$$

Radius of Available Wrench (rAw)

The radius of the available wrench (rAW) is the maximum magnitude of force that the system can generate uniformly in all directions while remaining within admissible tension limits [5]. Its optimization helps identify arrangements that simultaneously maximize force robustness, geometric and collision-avoidance constraints, resulting into an enlargement of the static workspace [5].

$$rAW = \min_i \left(\min \left(\frac{|d_{pi}|}{\|\mathbf{n}_{null,i}\|}, \frac{|d_{qi}|}{\|\mathbf{n}_{null,i}\|} \right) \right) \tag{2.25}$$

being $u_{null,i} = \text{null}(M_i)$ with M_i containing combinations of linearly independent cable unit vectors and d_{pi} and d_{qi} defined as:

$$\begin{aligned} d_{pi} &= \sum_{j \in I^+} \tau_{max} u_{nulli}^T u_j + \sum_{j \in I^-} \tau_{min} u_{nulli}^T u_j - m |u_{nulli}^T g| \\ d_{qi} &= - \sum_{j \in I^-} \tau_{max} u_{nulli}^T u_j - \sum_{j \in I^+} \tau_{min} u_{nulli}^T u_j - m |u_{nulli}^T g| \end{aligned} \tag{2.26}$$

However, since [5] consider the case of a point-mass payload, it does not take in consideration the torques, we may compute separately rAW$_{force}$ and rAW$_{moment}$, the maximum pure force (resp. torque) the platform can exert in any direction while generating zero net moment (resp. linear force).

Conditioning index

The conditioning index measures how uniformly the system can generate forces and moments in different directions. Mathematically:

$$\nu = \frac{\sigma_{\min}(J_p)}{\sigma_{\max}(J_p)} \in [0, 1] \tag{2.27}$$

$\sigma_{\min}$ and $\sigma_{\max}$ are the minimum and maximum singular values of the Jacobian matrix, obtained via Singular Value Decomposition (SVD) [6]. A small conditioning number indicates a well-conditioned system, far-away from singular configuration.

Manipulability

The manipulability index, defined as $w = \sqrt{\det(J_p J_p^\top)}$, measures the volume of the wrench ellipsoid. A value close to 0 means that the system hit a kinematic singularity, while large value means a large overall wrench generation capability.

2.4 Architecture Selection Strategy

Say that this is an option but then just announced which one we used, but it will be talken better in the next chapter

Once the feasibility constraints are satisfied, we may want to proceed either by

$$N = w_1\hat{\gamma} + w_2\,\mathrm{r\hat{A}w} + w_3\hat{\zeta} \tag{2.28}$$

where all the metrics are scaled to $[0, 1]$, or either choosing the solution that maximizes the radius of available wrench between all the solutions that satisfy $\gamma > \gamma_{\text{min}}$ and using the worst-case capacity margin as a tie breaker.

Our solution will finally be:

$$U^* = \arg\max_U (N(U)) \tag{2.29}$$

We are interested in working with an over-actuated system. Therefore, we will start using six cables for fully control the payload pose and three drones acting as an anti-gravitational force. A schematic is shown in Figure 2.7.

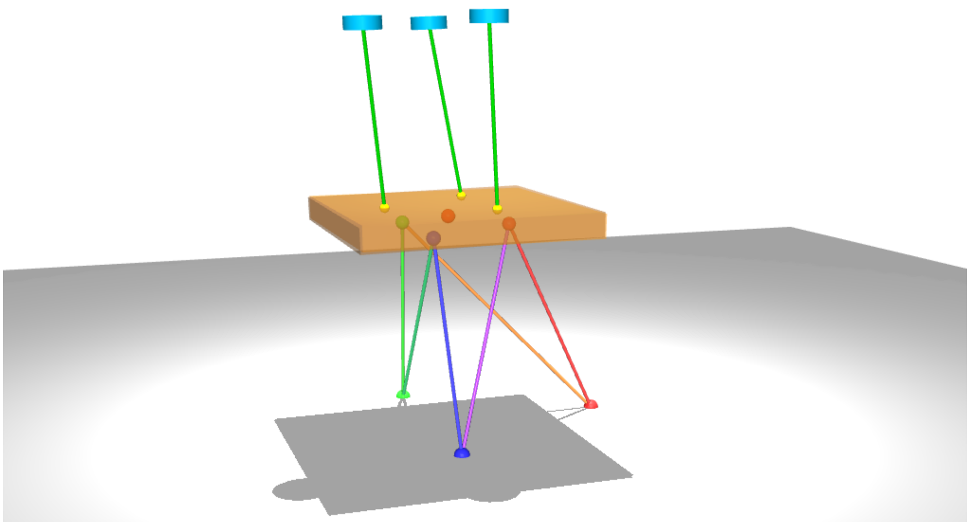

Figure 2.7: Over actuated ACTS with 6 ground cables and 3 UAVs

2.5 High level architecture

At high level, the architecture that enables the system to reach the goal is shown in Figure 2.8.

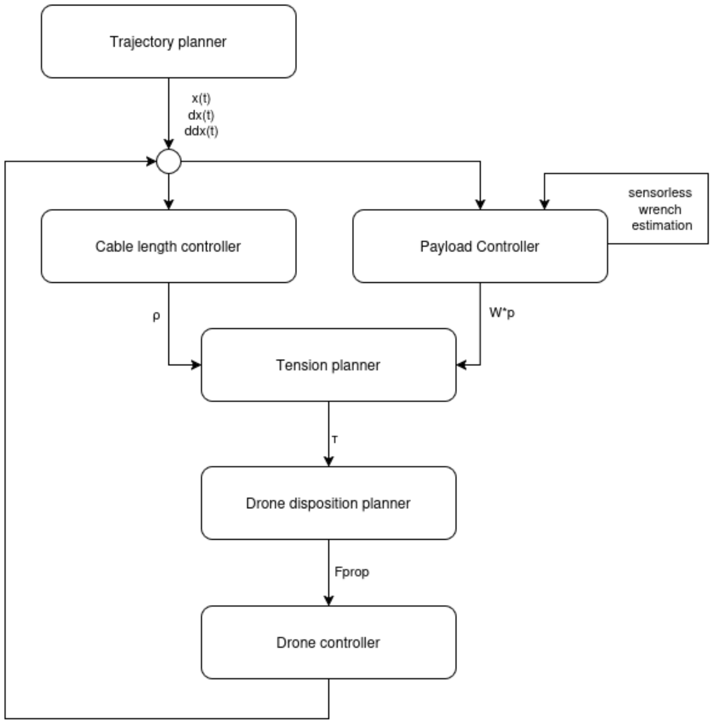

Figure 2.8: High level control scheme

Trajectory planner

The trajectory planner is able to compute the payload trajectory $x(t)$ and its velocity $\dot{x}(t)$ and acceleration $\ddot{x}(t)$ given the system's goal, such as:

1. Make the payload fly;
2. Bring the payload at the right height and orientation;
3. Approach the façade.

and/or the initial and final position of the payload.

Payload Controller

Then, the desired payload wrench W_p^* is computed for each point in the trajectory based on the previous state of the system. The desired payload wrench is computed from a geometric PD controller. The translational component is given by

$$F_p^* = m_p g + K_p(p^* - p) + K_d(\dot{p}^* - \dot{p}), \tag{2.30}$$

where p^* and p denote the desired and current payload positions, respectively.

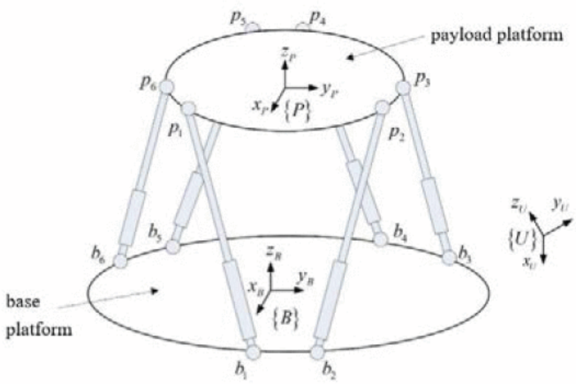

Figure 2.9: Gogh Stewart schematics [1]

For the rotational dynamics, the controller follows the geometric formulation on $SE(3)$ [7]. The attitude error and angular velocity error are defined as

$$\begin{aligned} e_R &= \frac{1}{2}\left(R_d^T R - R^T R_d\right)^\vee \\ e_\Omega &= \Omega - R^T R_d \Omega_d. \end{aligned} \tag{2.31}$$

where $(\cdot)^\vee$ denotes the inverse of the skew-symmetric (hat) operator.
Since the controller performs setpoint regulation, $\Omega_d = \mathbf{0}$, yielding $e_\Omega = \Omega$.
The desired control moment in the world frame is then

$$M_p^* = R(-K_R e_R + K_\Omega e_\Omega) \tag{2.32}$$

and the desired payload wrench becomes

$$W_p^* = \begin{bmatrix} F_p^* \\ M_p^* \end{bmatrix}. \tag{2.33}$$

Cable length controller (winch level control)

When transitioning from a desired payload pose to the next along the planned trajectory, the operational priority is given to the ground actuators, as they can be controllable with significantly higher precision than UAVs.

As the starting ACTS configuration consists of six ground cables, the ground subsystem can be modeled as a Gough-Stewart parallel platform (Figure 2.9), where the required positive cable tensions are guaranteed by the drones.

Given the desired payload position p^* and the rotation matrix ${}^0_pR^*$ that defines the desired orientation of the payload relative to the fixed world frame, the desired i-th ground cable vector ${}^0\mathbf{l}_i^*$ is geometrically defined as:

$${}^0\mathbf{l}_i^* = {}^0\mathbf{a}_i -{}^0 \boldsymbol{r}_i = {}^0\mathbf{a}_i - \left({}^0\mathbf{p}^* + {}^0_pR^* \, {}^p\mathbf{b}_i\right) \tag{2.34}$$

change this formula to go with the code or is it better change the code to go with this?

The target leg length is $\rho_i^* = \|l_i^*\|$. To ensure precise tracking of the ground cable lengths, we can either implement a PD controller at the winch level, where the velocity command $\dot{\rho}_i$ is given by:

$$\dot{\rho}_i = \dot{\rho}_i^* + K_i(\rho_i^* - \rho_i) + K_p(\dot{\rho}_i^* - \dot{\rho}_i) \tag{2.35}$$

Under the assumption of stationary winches ($\dot{a} = 0$), $\dot{\rho}^*_{ground} = -J_{p,ground}\xi^*$ (Equation 1.5), where ξ^* is the desired payload twist and $J_{p,ground} \in \mathbb{R}^{6\times6}$ is the Jacobian matrix mapping the payload twist to the ground cable length variation. In simulation, we implement a control directly on the cable length:

$$l_i = l_i^* + K_p(l_i^* - l_i) + K_d(\dot{l}_i^* - \dot{l}_i) \tag{2.36}$$

Tension planner

The tension planner is the layer responsible for guaranteeing that every cable stays taut throughout the motion. In fact, when working with ACTS, a payload wrench configuration is statically admissible if and only if there exists a set of strictly positive cable tensions ($\tau_i \geq \tau_{min} > 0$) that balance the target operational wrench. This is due to the fact that cables can only pull and not push. Mathematically, this layout optimization is formulated as:

$$\begin{aligned}
\min_{p_a} \quad & \sum_{i=1}^{n_a} \|F_{prop}(p_{a,i}, \tau_{optimal,i})\|^2 + K\|J_p\tau_{optimal} - W_p^*\|^2 \\
\text{subj. to} \quad & x_i^2 + y_i^2 + z_i^2 = l_i^2 \quad \forall i \in [1, n_a] \\
& \rho_{WCC}(\mathcal{U}) \leq 0 \\
& \rho_{IFC}(\mathcal{U}) \leq 0
\end{aligned} \tag{2.37}$$

which is a Nonlinear optimization Problem (NLP) due to the Euclidean distance constraints. In Section 2.5.1, we propose some ways to turn it into a Quadratic Programming problem (QP).

This objective function aims to minimize the propulsion force of the drones while satisfying the wrench equilibrium equations on the payload. The parameter K acts as a penality weight that prioritizes the tracking performance over energy efficiency; setting a high value (e.g., $K = 1000$), heavily penalizes any mismatch between the generated wrench and the commanded target wrench W_p^*.

For whatever positions p_a, the cable tensions τ are determined by a second inner loop

$$\begin{aligned}
\tau_{optimal} = \arg\min_{\tau} \; & \frac{1}{2}\tau^\top H \tau \\
\text{subj. to} \quad & J_p^\top(\mathbf{u})\tau = W_p^* \\
& \tau_{min,i} \leq \tau_i \leq \tau_{max,i} \quad \forall i \in [1, n_a + n_g]
\end{aligned} \tag{2.38}$$

The lower bound $\tau_{min,i} > 0$ in the wrench feasible condition is what enforces the taut-cable requirement.

As the winch-level control drives $\rho \to \rho^*$, tracking the payload's motion toward its target pose, and $\mathbf{u} \to \mathbf{u}^*$, the achievable wrench set used by the tension planner is itself changing over the trajectory, since it depends on $\mathbf{u}$. As a consequence, $\mathbf{W}_p^*$ may be reachable at the target configuration but momentarily unreachable at the actual configuration, with all tensions kept strictly positive. When this occurs, Equation 2.38 should relax the equality constraint $\mathbf{J}_p^\top(\mathbf{u})\,\tau = \mathbf{W}_p^*$ to a least-squares objective, while keeping $\tau_{min,i} > 0$ as a hard constraint.

$$\tau_{optimal} = \arg\min_{\tau} \left(\frac{1}{2}\tau^\top H \tau + \gamma \| J_p^\top(\mathbf{u})\tau - W_p^* \|^2 \right)$$
$$\text{subj. to} \quad \tau_{min,i} \leq \tau_i \leq t_{max,i} \quad \forall i \in [1, n_a + n_g] \tag{2.39}$$

This guarantees that the tension planner always returns a taut solution, one that approximates the desired wrench as closely as the actual geometry allows, rather than returning no solution at all during the transient.

For the initial stage of development, this general QP is replaced by a simplified strategy: every cable tension is fixed at the minimum admissible value,

$$\|\tau_i^*\| = \tau_{min}, \qquad \forall i \in [1, n_a + n_g]. \tag{2.40}$$

Drone disposition planner

Once the tension planner computes the scalar target tension magnitudes, the drones must dynamically adjust their orientation and propeller thrusts to impose these forces.

From the static equilibrium balance (Equation 1.13) the required propeller thrust vector is given by:

$$\mathbf{F}_{prop,i} = \mathbf{F}_{a,i} - \mathbf{F}_{g,i} = \mathbf{J}_{a,i}^\top \tau_i + m_i \mathbf{g} = T_i \, {}^0_i\mathbf{R}\, \mathbf{e}_3, \qquad T_i = \|\mathbf{F}_{prop,i}\| \tag{2.41}$$

which is then handed to the attitude controller, responsible for aligning the body z-axis with the commanded thrust vector.

2.5.1 Quadratic Approximation of the UAV Disposition Problem

delete it? We don't use it

The nonlinear optimization problem presented in the previous section:

$$\begin{aligned} \min_{p_a} \quad & \sum_{i=1}^{n_a} \|F_{prop}(p_{a,i}, \tau_{optimal})\| \\ \text{subj. to} \quad & x_i^2 + y_i^2 + z_i^2 = l_i^2 \quad \forall i \in [1, n_a] \\ & \rho_{WCC}(\mathcal{U}) \leq 0 \\ & \rho_{IFC}(\mathcal{U}) \leq 0 \end{aligned}$$

can be rewritten as a quadratic problem where possible.

Wrench closure condition

The rank condition $\text{rank}(J_p) = 6$ is not a smooth inequality. To overcome this problem, we could:

- Use the manipulability index squared $w^2 = \det(J_p J_p^\top) \geq w_{\min}^2 > 0$ in which $w = 0$ indicates a singular configuration;
- Impose positive value for the smaller singular value ϵ: $\sigma_{\min}(J_p) \geq \epsilon$

The cost function

The objective function can be approximated by replacing the sum of Euclidean norms with the sum of squared Euclidean norms,

$$\min_{p_a} \sum_{i=1}^{n_a} \|F_{prop}(p_{a,i}, \tau_{optimal})\| \approx \min_{p_a} \sum_{i=1}^{n_a} \|F_{prop}(p_{a,i}, \tau_{optimal})\|_2 = \min_{p_a} \sum_{i=1}^{n_a} F_{prop}^{\top} F_{prop} \tag{2.42}$$

as for definition $\|u\|_2 = \sqrt{u^{\top} u}$. This approximation penalizes large outliners more than small ones.

Therefore, the problem becomes:

$$\begin{aligned} \min_{p_a} \quad & \sum_{i=1}^{n_a} F_{prop}^{\top} F_{prop} \\ \text{subj. to} \quad & x_i^2 + y_i^2 + z_i^2 = l_i^2 \quad \forall i \in [1, n_a] \\ & w_{\text{min}}^2 - w^2 \leq 0 \\ & d_{\text{safe}} - d_{ij} \leq 0 \\ & d_{\text{safe}} - d_i \leq 0 \end{aligned} \tag{2.43}$$

Chapter 3
Experiments

How to structure this chapter:

1. Why did we chose to go for the method: controller and see if that pose is reacheable? [DONE]
2. Talk about Stewart configurations,
   - which are the indices values [DONE]
   - what these can tell us about everything [DONE]
   - How do I want each performance index? [DONE]
   - Show them in a graph
3. Given an UAV configuration (we chose triangle), analyze cable routing
   - Why we choose the triangle for the UAVs configuration? [DONE]
   - How the code work, how a given cable routing has been found [DONE]
   - How the results before simulation and after simulation differ?
   - Why we need to test it dynamically and not just trust the results we obtain before? (Answer below, write it better)
   - Rectangle-same really unstable configuration
   - line or diagonal results in not-feasible configurations, why? what does this mean? Geometric reasons?
4. Given the results of the previous step, lets analyze the first 3/5 configurations with all 6 possible uav configurations
   - Expetation: triangle being the best one, if not, which one is best? Rerun step 2 and see how things change
   - How drone attachment influences reaching the goal? Show evolution graph
5. Can a more spread-out hook configuration on the ground better the performance or it is irrelevant?
   - Does the answer change when the desired position is If so, it opens results to the moving hook point.further away?
   - Show example on the top configuration and a really bad one.

- Is there any papers/theorem that says so?
- Show evolution graph

6. How a configuration far away from the hook point on the ground performs?
   (a) Stewart: the position is reacheable with a relative small error, but it is not stable.
   (b) The hook position should cover more space on the ground
   (c) Is there a geometric reason/ theory for that?

3.1 Experimental methodology and evaluation framework

The objective of this chapter is to evaluate how the architectural design of an ACTS influences its ability to accurately and robustly manipulate a payload. The focus is on how different cable routing strategies, UAV attachment configurations, and ground hook placements influence the overall system performance.

In fact, a configuration is considered successful if the desired payload is reached with sufficiently small position and orientation errors, all cables remain under positive tension throughout the maneuver, and the payload converges to an equilibrium without persistent oscillations.

In this chapter we will answer questions like "Which configuration gives the best payload control performance according to the theoretical indices?" and "Which configuration would I actually choose for the robot?"

The geometric and static performance indices provided in Section 2.3 describe the instantaneous ability of a given architecture to generate payload wrenches and identify configurations that are close to kinematic singularities. However, these indices are evaluated under static or quasi-static assumptions and therefore cannot fully characterize the behavior of the closed-loop system during motion. During motion, cable directions, tensions and payload dynamics continuously evolve. Consequently, a configuration that appears favorable according to static indices may still exhibit oscillations, poor tracking or instability. Closed-loop simulation is therefore necessary to verify whether the theoretical advantages translate into practical performance.

All simulations are carried out using the MuJoCo (Multi-Joint Dynamics with Contact) physics extensions, allowing simulation of closed-loop systems, such as CDPRs and ACTSs, which is not possible in Gazebo. Another reason I switched from Gazebo to MuJoCo was the ability to model cables as tendons, which are minimum-path-length strings with wrapping and tracking, constraints [8]. However, for the purpose of this thesis, the cables are treated as pure physical constraints, without stiffness or dumping.

The payload mass, UAV model, controller gains, drone cable lengths, simulation duration, and desired payload pose are kept constant throughout all experiments to ensure a fair comparison between diferent architectural configurations. The only variables that change from one experiment to another are the cable routing, the UAV attachment geometry, and the placement of the ground hook points.

Two desired payload poses are taken in considerations. In the first scenario, the desired payload orientation coincides with the world reference frame; while in the other scenario a non-zero desired orientation is imposed. This allows to analyze separately the rotational motions from the translation tracking.

First, the classical Stewart configuration is analyzed and used as a reference architecture due to its geometric symmetry and extensive use in the literature. In fact, if even Stewart-like performs poorly, the problem is likely the controller rather than the architecture. The remaining experiments then investigate how modifications to the ground cable routing, UAV attachment geometry, and hook placement affect both the theoretical performance indices and the closed-loop behavior of the system. This progressive analysis makes it possible to identify which geometric features have the greatest influence on payload manipulation performance.

3.2 Stewart platform baseline

Thanks to its symmetrical geometry, the Stewart ground cable configuration is expected to ensure uniform torque generation in all directions while maintaining an optimal distance from singularities. Consequently, high values of the manipulability, capacity margin, radius of available wrench, and composite score are expected, together with a relatively high conditioning index. From a control perspective, these characteristics should result in accurate payload tracking while maintaining all cable tensions within their admissible limits.

3.2.1 Numerical Verification of the Implementation

Before comparing different ACTS architectures, one representative Stewart configuration is manually verified. The optimized cable tensions, payload wrench and UAV thrusts are recomputed directly from the governing equations presented in Chapter 2 and compared with the values returned by the implementation. This verification confirms the correctness of the numerical optimization and the controller implementation before performing the comparative study.

Configuration

Payload dimensions $2.0 \times 2.0 \times 0.2$
Desired pose $p^* = [0.5, -0.5, 2.0]^\top$
Desired orientation $R^* = \mathbb{I}$
Desired payload wrench $W_p^* = [0, 0, -9.21, 0, 0, 0]^\top$

attachment		hook points	
$A_4 \equiv A_5$	$[0.0, 1.1, 0.0]^\top$	$B_4 \equiv B_9$	$[0.5, 0.05, 1.9]^\top$
$A_6 \equiv A_7$	$[-0.9526, -0.55, 0.0]^\top$	$B_5 \equiv B_6$	$[-0.0237, -0.775, 1.9]^\top$
$A_8 \equiv A_9$	$[0.9526, -0.55, 0.0]^\top$	$B_7 \equiv B_8$	$[0.9763, -0.775, 1.9]^\top$

Using Equation 1.6b , the cable directions u_i are:

$$\begin{aligned} u_4 &= [-0.2245, 0.4713, -0.8529]^\top & u_7 &= [-0.7100, 0.0828, -0.6993]^\top \\ u_5 &= [-0.0089, 0.7024, -0.7117]^\top & u_8 &= [-0.0124, 0.1176, -0.9930]^\top \\ u_6 &= [-0.4545, 0.1047, -0.8846]^\top & u_9 &= [0.2215, -0.2937, -0.9299]^\top \end{aligned}$$

$$J_{p,ground} = \begin{bmatrix} -0,22445 & 0,47135 & -0,85291 & -0,42197 & 0,02245 & 0,12345 \\ -0,00888 & 0,70238 & -0,71175 & 0,26597 & -0,33812 & 0,33656 \\ -0,45452 & 0,10475 & -0,88456 & 0,25373 & -0,37586 & 0,29632 \\ -0,70998 & 0,08282 & -0,69934 & 0,20060 & 0,40409 & -0,52834 \\ -0,01239 & 0,11759 & -0,99298 & 0,28483 & 0,47420 & -0,47636 \\ 0,22151 & -0,29365 & -0,92989 & -0,54081 & -0,02215 & -0,12183 \end{bmatrix}$$

But I don't know where the drones may be, so what to do? I can do the same and compute also the cable vector for the drone cable and its respective jacobian

$$u_1 =$$
$$u_2 =$$
$$u_3 =$$

$$J_{p,drones} =] \; [$$

Payload wrench W_p

Cable tensions τ_i

Required drone thrust

Does it correspond

3.2.2 Comparison τ_{min} and τ_{opt}

As mentioned in Section 2.5, we evaluate two distinct tension allocation strategies:

- Minimal tension strategy, in which a minimum cable tension is imposed to each cable tension such that $\|\tau_i\| = \tau_{min}$.
- Optimal tension strategy, in which cable tensions are determined by the quadratic optimization problem in Equation 2.39

The performance indices and tracking errors under both strategies across three operational scenarios are summarized in Table 3.1.

Even though being faster, the minimal tension strategy has negative consequences on the system's robustness. By locking all cable tensions to their absolute lowest limit ($\tau_{\min}$), the range of available tension choices shrinks to a single point. As a result, the controller loses its ability to adjust forces, reducing the payload's Available Wrench Set (AWS) to just a single fixed outcome. [9] As a result, any deviation of the desired payload wrench W_p^* from this single point results in a negative capacity margin (as seen in Scenario A where $\gamma = -0.65562$). This indicates that the system is theoretically incapable of resisting external disturbances or maintaining equilibrium without violating the tension constraints.

Switching to the optimal tension allocation, allows the cable tensions to vary dynamically between τ_{min} and τ_{max}. As a result, the radius of the Available Wrench increases and the capacity margin takes on a positive value ($\gamma = 3.74947$). Thus, also the composite score increases. Differently, changing the tension allocation algorithm has no effect on the conditioning index or manipulability because they are kinematic metrics that only depend on the platform's geometric position.

Give examples

Because this study focuses on offline structural and kinematic analysis rather than real-time control, the extra computation time does not compromise our findings. For future real-time applications, this hierarchical formulation can be mapped to more efficient schemes.

show how position and orientation error changes overtime

On the other hand, it is worth to notice that, since the objective function does not contain any state-tracking terms, this optimizer does not directly minimize the payload's real-time position or orientation tracking errors. Because of this, a spatial configuration that yields globally minimal propulsion effort does not guarantee a minimum in tracking error.

Metric / Parameter	**Scenario A** $\tau_{\min}$	**Scenario A** τ_{opt}	**Scenario B** $\tau_{\min}$	**Scenario B** τ_{opt}
Desired Position	$[0.5, -0.5, 2.0]^\top$		$[0.5, -0.5, 2.0]^\top$	
Desired Orientation	$[1.0, 0.0, 0.0, 0.0]^\top$		$[1.0, 0.1, 0.1, 0.1]^\top$	
Conditioning Index	0.10256	0.09621	0.07639	0.06621
Manipulability	0.47759	0.43684	0.36192	0.30017
Radius Avail. Wrench	16.46686	19.52504	18.66125	20.29844
Capacity Margin	-0.65562	3.74947	-1.93639	2.28516
Worst-case CM	0.86233	0.63971	0.86253	0.67165
Composite Score	5.55786	7.97141	5.86246	7.75175
Position Error	0.00209	0.00285	0.00183	0.00271
Orientation Error	0.00125	0.00200	0.00127	0.00299

Table 3.1: Comparative performance indices and tracking errors for Stewart baseline configuration across Scenarios A and B.

Although the Stewart configuration exhibits good overall performance and provides an excellent reference architecture, the proposed ACTS differs from a classical Stewart platform because the UAV cables can be connected to the payload in several different ways. Moreover, different ground cable routings may further improve specific performance aspects such as stability, wrench generation capability, or robustness. For these reasons, the Stewart configuration is adopted as a baseline against which all alternative architectures are compared.

An ideal ACTS architecture should remain far from singular configurations, generate payload wrenches efficiently in every direction, and maintain sufficient redundancy to accommodate disturbances while respecting cable tension limits. Consequently, all the considered performance indices should be as big as possible.

The considerations from now on will be done using the optimal tension strategy.

3.3 Ground cable routing

Once the controller has been validated using the Stewart baseline, we can proceed to consider different ways to connect six ground cables to the platform (Figure 3.2) and the drone (Figure 3.3). The same ground configuration hook-anchor can change the system's stability based on which hook is connected to which anchor point on the ground. Therefore, we will start considering the drone attachment configuration as fixed to the triangular arrangement and vary only the routing of the six ground cables. This specific UAV configuration is chosen because it provides a symmetric distribution of the three drones around the payload. This approach allows one to isolate the influence of the ground cable routing and attribute differences in performance solely to the ground architecture. Finally, in Section 3.5, the impact of various UAV attachment geometries is examined independently.

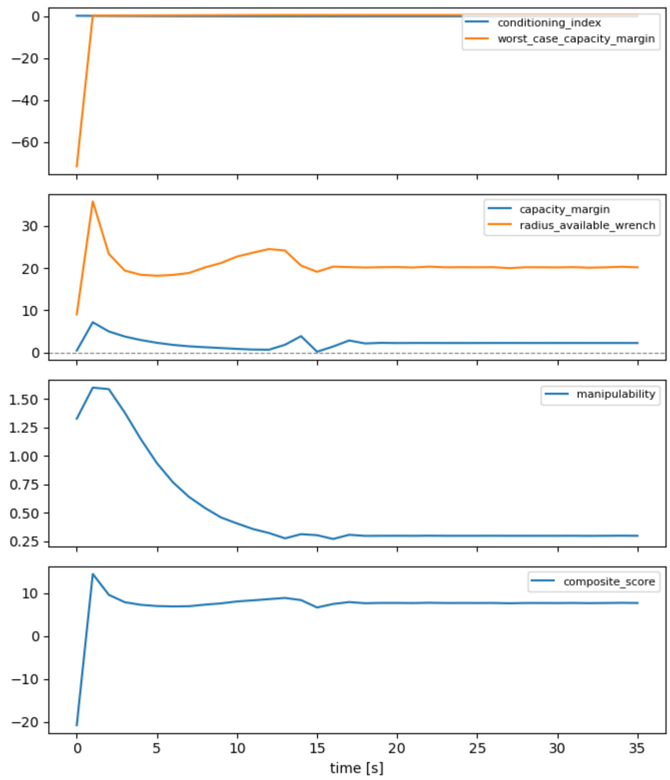

Figure 3.1: Performance indices for

Focusing on the gound cable routing, there are at most $6! = 720$ physically distinct valid ways to route the six ground cables. Scoring an architecture using only one arbitrarly chosen routing raise the question on how lucky the routing choise was and how good the architecture actually is.

To eliminate this source of bias, an exhaustive routing search is performed for every pair of payload and ground layouts. All physically valid routings are enumerated while respecting the maximum cable capacity of each attachment point. For every routing, the corresponding payload Jacobian J_p is constructed according to Equation 1.7, and the routing is evaluated over a representative set of payload poses. At each pose, the routing must satisfy the necessary conditions introduced in Section 2.3. Routings that fail any of these conditions are discarded. Among the surviving candidates, the routing is assigned its worst-case performance across all poses, and the architecture is finally characterized by the routing that maximizes the conditioning index. In this way, architectures are compared according to their best achievable routing rather than to an arbitrary cable assignment. Architectures are subsequently ranked against one another by worst-case conditioning index.

The composite score N is not used for ranking routings during the exhaustive search because it requires information that is only available after solving the tension planner, such as the optimal cable tensions and the available wrench set. Computing its capacity-based terms is also computationally expensive and unreliable near singular configurations. Additionally, its weighting factors have not been experimentally validated, making it unsuitable for selecting the best routing. Instead, the search uses the conditioning index (with manipulability as a supporting metric), as it is a parameter-free, geometry-based measure that can be efficiently computed directly from the Jacobian. The composite score

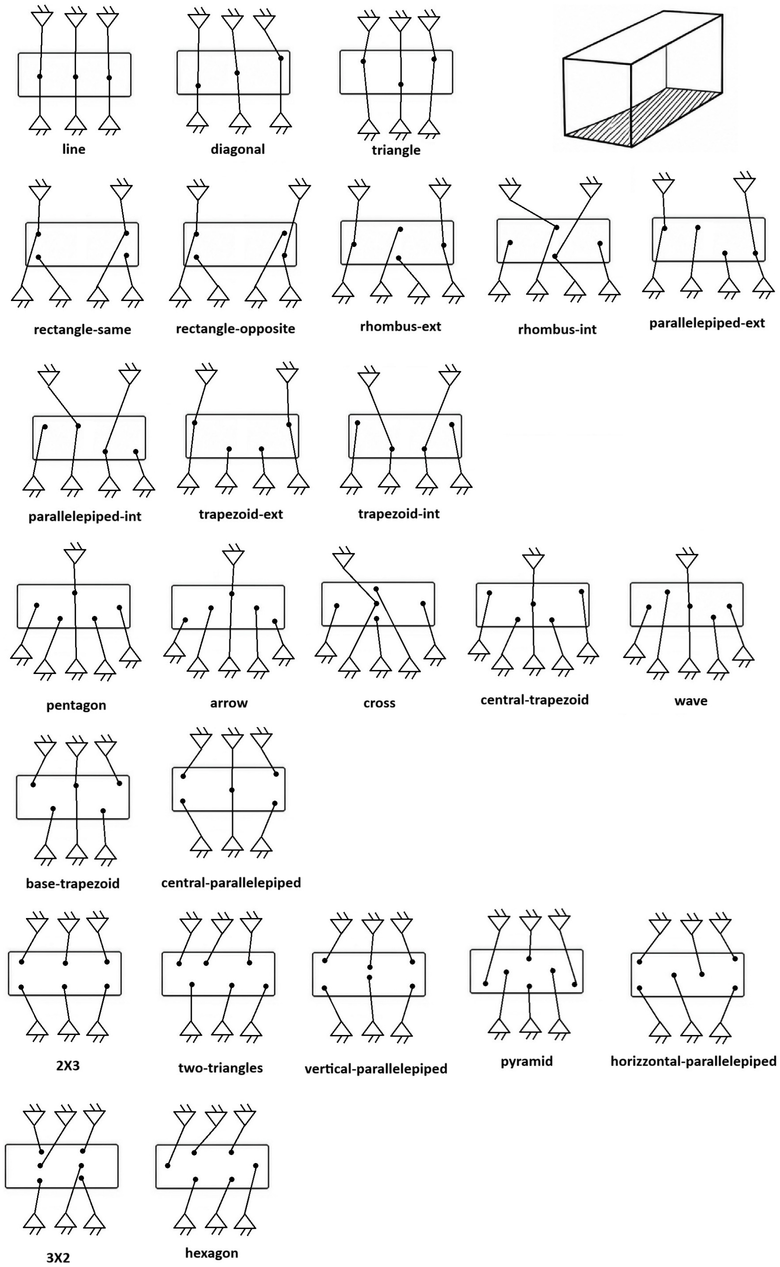

Figure 3.2: Possible 6 UGVs connections

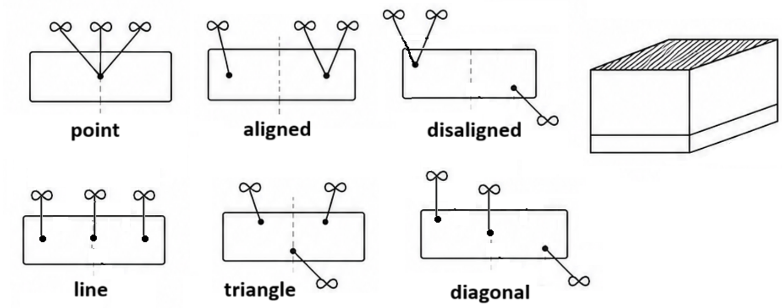

Figure 3.3: Possible 3 UAVs connection with one cable

and capacity-based metrics are therefore evaluated later during closed-loop simulations, once a routing and tension solution are available.

Algorithm 1 Cable Routing Optimization and Architecture Ranking

for all (payload_layout, ground_layout) pairs **do**
    Enumerate every valid 6-cable routing
    for all routing **do**
        for all pose $\in$ test-pose set **do**
            Build J_p ▷ Eq. 1.7
            Check Wrench Closure ▷ $\text{rank}(J_p) = 6$
            Check Wrench Feasibility ▷ $0 \leq \tau_{\min} \leq \tau \leq \tau_{\max}$
            Check Interference-Freedom ▷ Eq. 2.20
        end for
        Record this routing's **worst-case** score across all poses
    end for
    Keep the **best** surviving routing for this architecture
end for

Rank all architectures by their best routing's worst-case conditioning index Generate MuJoCo XML for the top 5

How can I know a configuration is mure suscettibile to be singular? Something with a jacobian?

How the results before simulation and after simulation differ?

The theoretical indices assume: static equilibrium, ideal cables, no inertia, no controller dynamics. The simulation includes: payload inertia, controller dynamics, coupling, transient motion, oscillations.

Update with correct datas

Pay. Layout	**Gnd. Layout**	$N_{\textbf{rout}}$	$N_{\textbf{pos}}$	**Feas.**	ν	w	$\sigma_{\min}$	$\sigma_{\max}$	**Rank**	
2X3	pentagon	360	8	✓	0.1402	0.3680	0.3275	2.3452	✓	A-E—
two-triangles	pentagon	360	8	✓	0.1383	0.2169	0.3241	2.3566	✓	A-E—
pentagon	2X3	360	8	✓	0.1354	0.3773	0.3169	2.3401	✓	A-A—
hexagon	pyramid	720	8	✓	0.1343	0.1664	0.3180	2.3794	✓	A-D—
pentagon	pentagon	168	8	✓	0.1316	0.2494	0.2899	2.3420	✓	A-E—

Table 3.2: Ground Cable Routing and Layout Performance Analysis

Rectangle-same really unstable configuration

The theoretical indices evaluate the intrinsic geometric and static properties of a cable routing. They identify configurations that are close to kinematic or wrench singularities and rank architectures according to their ability to generate payload wrenches. However, they cannot predict the transient response of the closed-loop system. Dynamic effects such as payload inertia, controller gains, load redistribution and restoring moments are only captured by simulation. Consequently, while poor theoretical indices almost always indicate poor dynamic performance, the converse is not necessarily true: a routing with good static indices may still exhibit oscillations or poor tracking due to dynamic phenomena.

This is exactly what your Rectangle-Same example demonstrates: the Jacobian is full rank and the static indices are acceptable, yet the payload rocks because of an uneven load distribution and poor torsional stiffness. That example provides strong evidence that the theoretical indices are necessary but not sufficient for predicting closed-loop behavior.

line or diagonal results in not-feasible configurations, why? what does this mean? Geometric reasons?

The line and diagonal configurations approach kinematic or wrench singularities, where the cable arrangement cannot generate the desired six-dimensional wrench while maintaining positive tensions. Conversely, the Rectangle-Same configuration remains kinematically feasible but exhibits poor rotational stiffness due to its asymmetric cable pairing. As a result, although the Jacobian remains full rank, the payload develops a rocking motion that cannot be predicted by the geometric performance indices alone.

talk about triangle-triangle combination

Other combinations that give interesting results?

3.4 Ground anchor placement

Increasing the spacing between the ground hooks enlarges the available wrench set and moves the system farther from wrench singularities, explaining the observed improvement in tracking performance.

3.5 Drone hook placement

Add link to the repository

A full-rank Jacobian is necessary but not sufficient for good closed-loop performance.

The Jacobian says: "Can I generate arbitrary infinitesimal wrenches?", and not "Will the closed-loop system remain dynamically stable?"

Your rectangle is the perfect example. It has full rank, acceptable manipulability, acceptable conditioning, yet poor torsional stiffness, uneven load distribution, asymmetric restoring moments, oscillatory dynamics

In other words: The Jacobian is only a local kinematic property. The simulation captures: inertia, controller, load redistribution, dynamic coupling, restoring torques, which the Jacobian completely ignores.

Chapter 4

Conclusions

Future developments

- Consider different configuration rather then 6 on the ground and 3 on the top
- Façade interaction

Also answering these questions:

- Is it better having the drone connected through one or two cables to the payload in terms of easy maneuverability, feasible workspace, available wrench set and energy consumption?
- How accuracy on drones' position affects the accuracy on the platform's position?
- Are UAVs (resp. UGVs) strictly connected to the upper (resp. down) face or can they be attached to the down (resp. upper) face?

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Appendix A

Singularities and Performance Indices

write it better

Cable-driven parallel robots (CDPRs) and aerial cable-driven systems experience critical operating limits where force transmission or controllability degrades. Table A.1 summarizes these operational singularities and their physical consequences.

Type	**Cause**	**Detection Index**	**Consequence**
Kinematic	$\mathrm{rank}(J_p) < 6$	Condition Number, Manipulability	Loss of platform Degrees of Freedom (DoF)
Wrench	Desired wrench outside AWS	Capacity Margin, RAW	Target wrench cannot be generated
Tension	Negative or zero cable tensions	Tension optimization	Cable slack / loss of constraint
Stiffness	Poor cable distribution	Stiffness Matrix, Simulation	Unwanted oscillations and rocking
Dynamic	Closed-loop instability	Eigenvalues, Simulation	Trajectory tracking failure

Table A.1: Classification of Singularity Types in Cable-Driven Parallel Systems